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Project Synopsis

On

1. Abuse Detection using AI

2. AI Resume Screening

AI Interview Screening

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ARTIFICIAL INTELEGENCE AND MACHIN LEARNING
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(A Constituent College of Sri Siddhartha Academy of Higher Education)

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Abuse Detection using AI

Description:

Abuse Detection using Artificial Intelligence is a technology-driven approach designed to identify harmful, toxic, or abusive content in digital communication platforms such as social media, chat applications, online gaming, forums, and email systems. In today's digital world, where millions of messages are shared every second, manually monitoring content has become nearly impossible. AI steps in as a smart assistant that can read, understand, and evaluate text in real time to protect users from online harassment and cyberbullying.

The core idea behind abuse detection is to train machine learning models to recognize patterns of harmful language. These models learn from large datasets containing examples of abusive and non-abusive messages. By analyzing words, sentence structure, context, and emotional tone, AI systems can classify content into categories such as toxic, offensive, hate speech, threatening, or safe. Natural Language Processing (NLP) techniques play a key role in helping the system understand human language beyond simple keyword matching. Instead of only detecting specific banned words, modern AI systems understand context, sarcasm, and intent to a certain extent. For example, a simple keyword-based system may flag a harmless joke because it contains a sensitive word, whereas an AI-based system evaluates the overall meaning before making a decision. Advanced models such as deep learning neural networks improve accuracy by learning complex language patterns. These systems can continuously improve as they are exposed to new data.

Abuse detection systems are commonly used by social media platforms to automatically filter comments, warn users before posting offensive messages, or temporarily block harmful accounts. They help create safer digital environments, especially for children and teenagers who are more vulnerable to online abuse. In corporate environments, AI-based monitoring tools also help maintain professional communication standards.

However, building an effective abuse detection system comes with challenges. Language is complex and constantly evolving. Slang, coded language, sarcasm, and cultural differences make detection difficult. There is also a risk of bias if the training data is not diverse, which may result in unfair targeting of certain communities. Therefore, ethical AI development and regular model evaluation are extremely important.

From a technical perspective, the system usually involves data collection, text preprocessing (removing punctuation, stopwords, and normalization), feature extraction (such as TF-IDF or word embeddings), model training (using algorithms like Logistic Regression, Naive Bayes, or deep learning models), and performance evaluation using metrics like accuracy, precision, recall, and F1-score.

In conclusion, Abuse Detection using AI is not just a technical innovation but a social necessity in the digital age. It combines machine learning and human-centered design to reduce harm and promote respectful online communication. While AI cannot completely replace human moderation, it significantly reduces workload and acts as a first line of defense against harmful content, helping build a safer and more inclusive online community.

AI Resume Screening

Description:

AI Resume Screening is an intelligent recruitment support system that uses Artificial Intelligence to automatically analyze, filter, and rank job applications. In today's competitive job market, companies often receive hundreds or even thousands of resumes for a single job opening. Manually reviewing each resume is time-consuming, costly, and sometimes inconsistent. AI-based resume screening helps recruiters quickly identify the most suitable candidates by evaluating resumes based on predefined criteria such as skills, experience, education, certifications, and job relevance.

At the heart of AI resume screening lies **Machine Learning (ML)** and **Natural Language Processing (NLP)**. Resumes are typically unstructured documents written in different formats and styles. NLP techniques allow the system to read and understand the content of resumes just like a human recruiter would. The system extracts key information such as technical skills, work experience, job titles, years of experience, and academic qualifications. It then compares this information with the job description to determine how well a candidate matches the role. For example, if a company is hiring for a Python Developer role, the AI system checks for keywords and contextual relevance related to Python programming, frameworks (like Django or Flask), databases, and project experience. However, unlike simple keyword matching systems, advanced AI models understand context. A candidate who has "developed machine learning models using Python" would rank higher than someone who merely lists "Python" without practical experience. The system works in several stages. First, resumes are collected and converted into machine-readable text. Next comes preprocessing, which includes removing unnecessary characters, standardizing text, and tokenization. Then, feature extraction techniques such as TF-IDF or word embeddings convert text into numerical representations. A machine learning model—such as Logistic Regression, Random Forest, or even deep learning models—is trained on historical hiring data to predict candidate suitability. Finally, the system ranks candidates and may even generate a suitability score.

One major advantage of AI resume screening is efficiency. It significantly reduces the time required for shortlisting candidates and helps recruiters focus on high-potential applicants. It also promotes consistency by applying the same evaluation criteria to every candidate. In large organizations, this leads to faster hiring cycles and reduced recruitment costs.

However, ethical considerations are extremely important. If the AI model is trained on biased historical data, it may unintentionally favor certain groups over others. Therefore, transparency, fairness testing, and continuous monitoring are necessary to ensure unbiased hiring practices. AI should assist recruiters—not replace human judgment entirely.

In conclusion, AI Resume Screening is a powerful application of Artificial Intelligence in Human Resource Management. It combines automation, data analysis, and intelligent decision-making to streamline recruitment processes. When designed responsibly, it enhances efficiency, reduces human workload, and supports fair and data-driven hiring decisions, making it a valuable tool in modern talent acquisition systems.

AI Interview Screening

Description:

AI Interview Screening is an advanced recruitment technology that uses Artificial Intelligence to evaluate candidates during the interview process. Unlike traditional resume screening, this system analyzes a candidate's performance in video, audio, or text-based interviews to assess communication skills, confidence, technical knowledge, emotional tone, and overall suitability for a job role. In today's fast-paced hiring environment, organizations use AI interview screening to streamline recruitment while maintaining consistency and efficiency.

In conventional interviews, human recruiters evaluate candidates based on observation, questioning, and personal judgment. However, human evaluation may sometimes be influenced by fatigue, unconscious bias, or inconsistency. AI interview screening systems assist by providing data-driven insights. These systems use Machine Learning (ML), Natural Language Processing (NLP), and sometimes Computer Vision to analyze responses in a structured and measurable way.

For video interviews, AI tools can analyze facial expressions, eye contact, speech clarity, tone of voice, and body language. For audio interviews, speech recognition models convert spoken words into text and evaluate vocabulary, fluency, and confidence. In text-based interviews, NLP techniques assess grammar, keyword relevance, and response quality. The system compares candidate responses with predefined job requirements and ideal answer patterns to generate a performance score.

The working process of AI interview screening generally includes several stages. First, interview responses are recorded and converted into structured data. Next, preprocessing techniques clean the data by removing noise and irrelevant elements. Then, feature extraction methods identify key indicators such as sentiment, keyword usage, speaking pace, and emotional tone. Machine learning models, trained on previous interview data, predict the candidate's suitability score. Finally, the system generates a ranked list of candidates or detailed evaluation reports for recruiters.

One major advantage of AI interview screening is efficiency. Companies can conduct large-scale hiring without scheduling multiple human interview panels. It ensures standardized evaluation criteria for all candidates and reduces the initial screening workload. Additionally, AI systems can operate 24/7, allowing candidates to complete interviews at their convenience.

However, ethical considerations are critical in AI-driven interviews. Concerns include privacy, data security, algorithmic bias, and transparency. If training data contains bias, the system may unfairly disadvantage certain groups. Therefore, AI interview screening should be designed responsibly, regularly audited, and used as a support tool rather than a replacement for human decision-making.

In conclusion, AI Interview Screening represents the integration of Artificial Intelligence into modern recruitment practices. It enhances hiring efficiency, provides structured candidate evaluation, and supports data-driven decision-making. When implemented ethically and transparently, it becomes a powerful tool that assists recruiters in identifying the best talent while maintaining fairness and objectivity.